

EDUCATION

PhD, Electrical and Computer Engineering, University of Delaware, Delaware, USA	3.5/4.0	2025 — Present
B.Tech, Electrical and Electronics Engineering, IIT, Bhubaneswar, India	7.8/10.0	2021 — 2025
Senior Secondary School (XII), Chinmaya Vidyalaya, Delhi, India	90.2%	2020 — 2021
Secondary School (X), Chinmaya Vidyalaya, Delhi, India	96.5%	2018 — 2019

PHD RESEARCH FOCUS

Machine Learning for Hardware Security & Computer Systems Aug 2025 — Present

- Conducting research on leveraging machine learning methodologies to enhance hardware security and fortify computer system architectures.
- Exploring ML-driven threat detection, hardware Trojan mitigation, and resilient system design.
- Tech Stack: Python, PyTorch, C/C++, Verilog, Linux

RESEARCH EXPERIENCE

PhD Fellow Aug 2025 — Present

University of Delaware
Newark, DE, USA

- Investigating machine learning applications to enhance hardware security and computer networks.

Research Intern Jan 2025 — May 2025

Indian Institute of Technology, Kharagpur
Remote, India

- Researched episodic meta-learning for histopathology image classification, evaluating CNNs against Prototypical Networks under Prof. Nikita Saxena. (Tech Stack: Python, PyTorch, Few-Shot Meta-Learning)

Summer Research Intern Mar 2024 — Dec 2024

Johns Hopkins University
Remote, MD, USA

- Implemented DeepONets, Physics-Informed Neural Networks (PINNs), and Transformers using the JAX framework under Prof. Somdatta Goswami. (Tech Stack: JAX, Python, DeepONets, PINNs, Transformers)

Summer School Fellow Jul 2024

University of Zurich
Zurich, Switzerland

- Completed "Deep Dive into Blockchain," analyzing consensus mechanisms, staking, and AMMs across platforms like Hedera, Cardano, and Polkadot.

Summer Research Intern May 2023 — Jul 2023

Indian Institute of Technology, Delhi
New Delhi, India

- Investigated synthetic data augmentation for whole-slide histopathology image classification using GANs and Diffusion models under Prof. Anurag S. Rathore. (Tech Stack: Python, TensorFlow, PyTorch, GANs, Diffusion Models)

Summer Research Intern May 2022 — Jul 2022

Indian Institute of Technology, Indore
Indore, India

- Developed Natural Language Processing (NLP) tools for sentiment analysis and text classification under Prof. Vimal Bhatia. (Tech Stack: Python, NLTK, Scikit-Learn)

PUBLICATIONS

- Journal:** Balraj, K., Ramteke, M., **Dash, A.**, Rathore, A.S. (2026). Exploring the impact of data augmentation on whole-slide image classification: A comparative analysis between original medical datasets and fully synthetic images. *Multimedia Tools and Applications*, 85, 305. DOI: 10.1007/s11042-026-21421-4
- Manuscript- In- Communication:** **Dash, A.**, Saha, R., Saxena, N. Episodic Meta-Learning for Histopathology: CNNs vs Prototypical Networks. *Communicated to Computers in Biology and Medicine*.
- Conference:** Mishra, D.P., **Dash, A.**, Senapati, R. (2023). Vehicle-to-grid & grid-to-vehicle power quality improvement. *IEEE 3rd STPEC*, pp. 1–5. DOI: 10.1109/STPEC.2023.10431016
- Book Chapter:** Mishra, D.P., **Dash, A.**, Kumar, A., Salkuti, S.R. (2025). AI-powered strategies for efficient EV energy management. In *AI for Integrated Smart Energy Systems in EVs*, Springer, Cham.
- Book Chapter:** Mishra, D.P., **Dash, A.**, Kumar, A., Salkuti, S.R. (2025). AI-driven innovations in EV energy management. In *AI for Integrated Smart Energy Systems in EVs*, Springer, Cham.

PROJECTS

4-bit Carry Look-Ahead Adder & Custom VLSI Design | Cadence Virtuoso [GitHub]

- Implemented a full custom, transistor-level 4-bit Carry Look-Ahead Adder (CLA).
- Executed end-to-end custom IC design flow: schematic capture, symbol design, transient/DC parametric simulations, layout placement, DRC and LVS verification.
- Tech Stack: Cadence Virtuoso, Spectre Simulator, CMOS VLSI, DRC/LVS Verification

- Optimization Dynamics for Trojan Network Detection | Machine Learning [\[GitHub\]](#)
- Formulated and evaluated L2-regularized logistic regression models in Julia to classify malicious Trojan traffic.
 - Implemented and benchmarked first-order optimization algorithms (GD, SGD, Nesterov/Momentum, Adam) to analyze convergence rates, loss trajectories and step-size sensitivity.
 - Tech Stack: Julia, Pluto.jl, Optimization Theory, Logistic Regression, Linear Algebra
- PM 2.5 Level Prediction Using Weather Data | D³ Hackathon (3rd Place) [\[GitHub\]](#)
- Developed and deployed a machine learning model using historical weather data to predict PM 2.5 levels for environmental monitoring.
 - Tech Stack: Python, Scikit-Learn, Streamlit

ACHIEVEMENTS

- University of Delaware, USA** – Selected for Department fellowship for Fall Semester, Sept – Dec, 2025.
- University of Zurich, Switzerland** – Selected for a fully funded scholarship to attend the UZH International Summer School “Deep Dive into Blockchain”, July 1–19, 2024.

SKILLS

- Core:** Machine Learning, Hardware Security, Computer Systems & Networking, Deep Learning, NLP
- Languages:** Python, C/C++, Verilog, VHDL, Julia, HTML, CSS, MATLAB
- Frameworks & Libraries:** PyTorch, TensorFlow, JAX, Keras, NLTK, Scikit-Learn
- Tools & OS:** Git, GitHub, Visual Studio, CAD, Linux, Windows, macOS
- Management:** Technical communication, Collaborative research

POSITIONS OF RESPONSIBILITY

Google Developer Student Club (GDSC) Lead	IIIT Bhubaneswar	2023 — 2024
IEEE Student Branch Secretary	IIIT Bhubaneswar	2023 — 2025
Microsoft Learn Student Ambassador	IIIT Bhubaneswar	2022 — 2025